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Quiz 4 Bonus

CHE 311

1.

$$V_2 = \sum_{n=1}^{\infty} \sum_{m=1}^{\infty} C_{nm} \cos\left(\frac{2n-1}{2a} \pi x\right) \cos\left(\frac{2m-1}{2b} \pi y\right)$$

$$\begin{aligned} & \sum_{n=1}^{\infty} \sum_{m=1}^{\infty} C_{nm} \left(\frac{2n-1}{2a} \pi\right)^2 \cos\left(\frac{2n-1}{2a} \pi x\right) \cos\left(\frac{2m-1}{2b} \pi y\right) \\ & + \sum_{n=1}^{\infty} \sum_{m=1}^{\infty} C_{nm} \left(\frac{2m-1}{2b} \pi\right)^2 \cos\left(\frac{2n-1}{2a} \pi x\right) \cos\left(\frac{2m-1}{2b} \pi y\right) \\ & = -\frac{\Delta \mathcal{P}}{ML} \end{aligned}$$

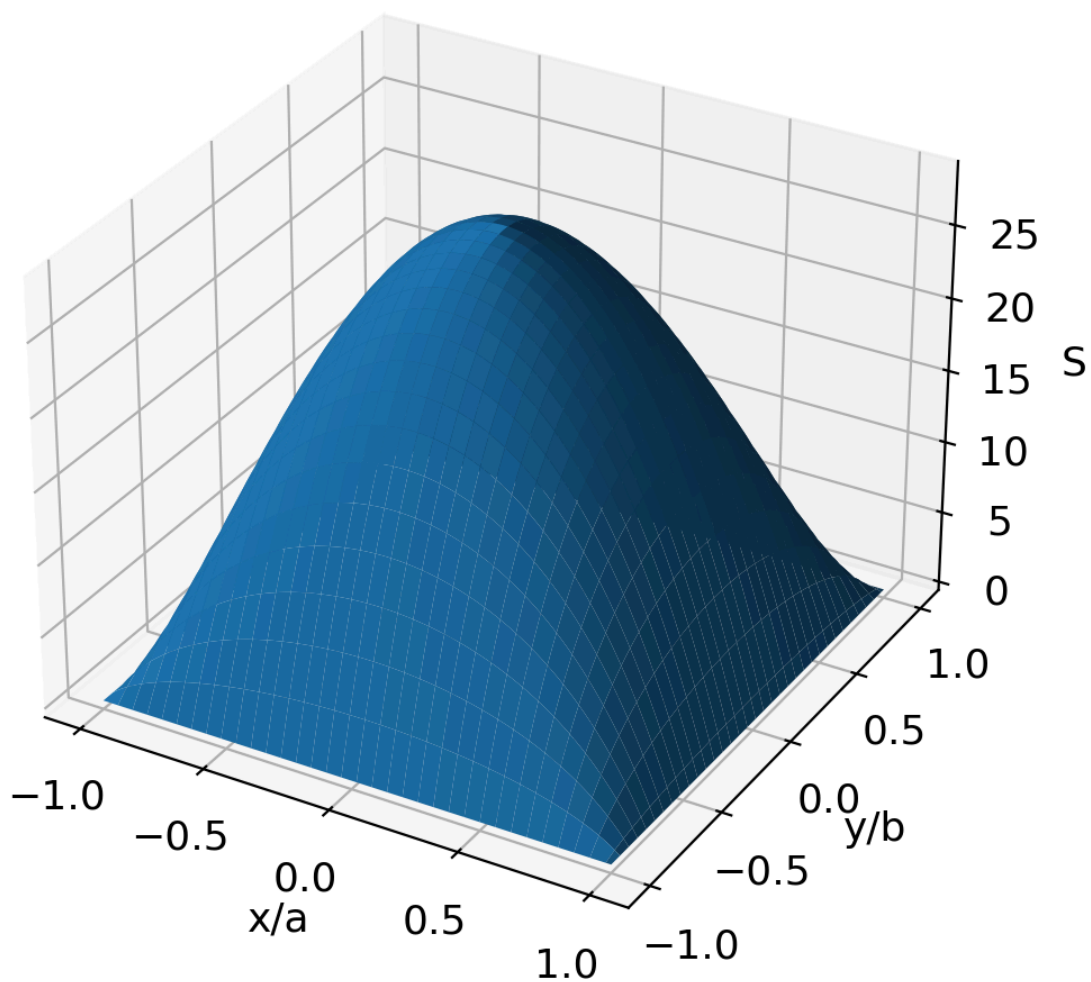
$$\frac{a}{\pi} \int_{-\frac{\pi}{2a}}^{\frac{\pi}{2a}} \cos(n\pi x) \cos(m\pi x) dx = \delta_{nm}$$

$$\begin{aligned} & C_{nm} \left[\left(\frac{2n-1}{2a} \pi\right)^2 + \left(\frac{2m-1}{2b} \pi\right)^2 \right] \\ & = -\frac{\Delta \mathcal{P}}{MLab} \int_{-a}^{+a} dx \cos\left(\frac{2n-1}{2a} \pi x\right) \int_{-b}^{+b} dy \cos\left(\frac{2m-1}{2b} \pi y\right) \end{aligned}$$

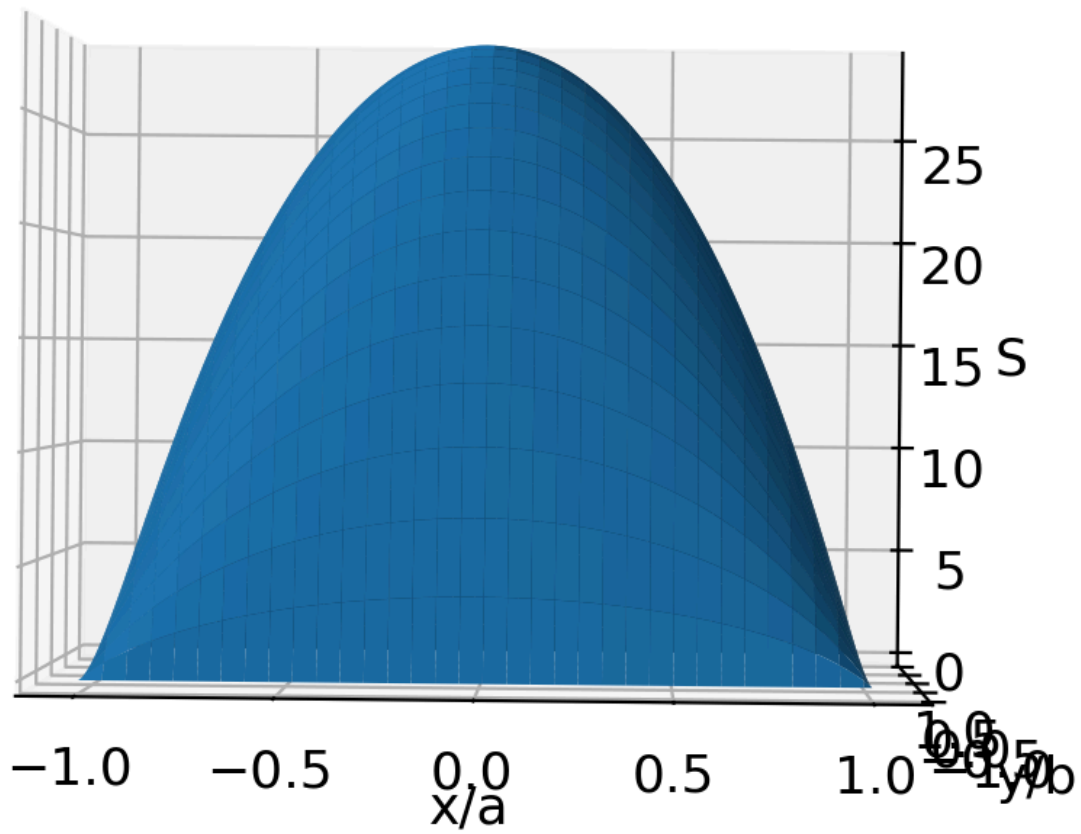
$$= -\frac{16 \Delta \mathcal{P} (-1)^{n+m}}{ML(2n-1)(2m-1)\pi^2}$$

$$\rightarrow V_2 = -\frac{64 \Delta \mathcal{P} a^2}{ML \pi^4} \sum_{n=1}^{\infty} \sum_{m=1}^{\infty} \frac{(-1)^{n+m}}{(2n-1)(2m-1) \left[(2n-1)^2 + \left(\frac{a}{b}\right)^2 (2m-1)^2 \right]} \cdot \cos\left(\frac{2n-1}{2a} \pi x\right) \cos\left(\frac{2m-1}{2b} \pi y\right)$$

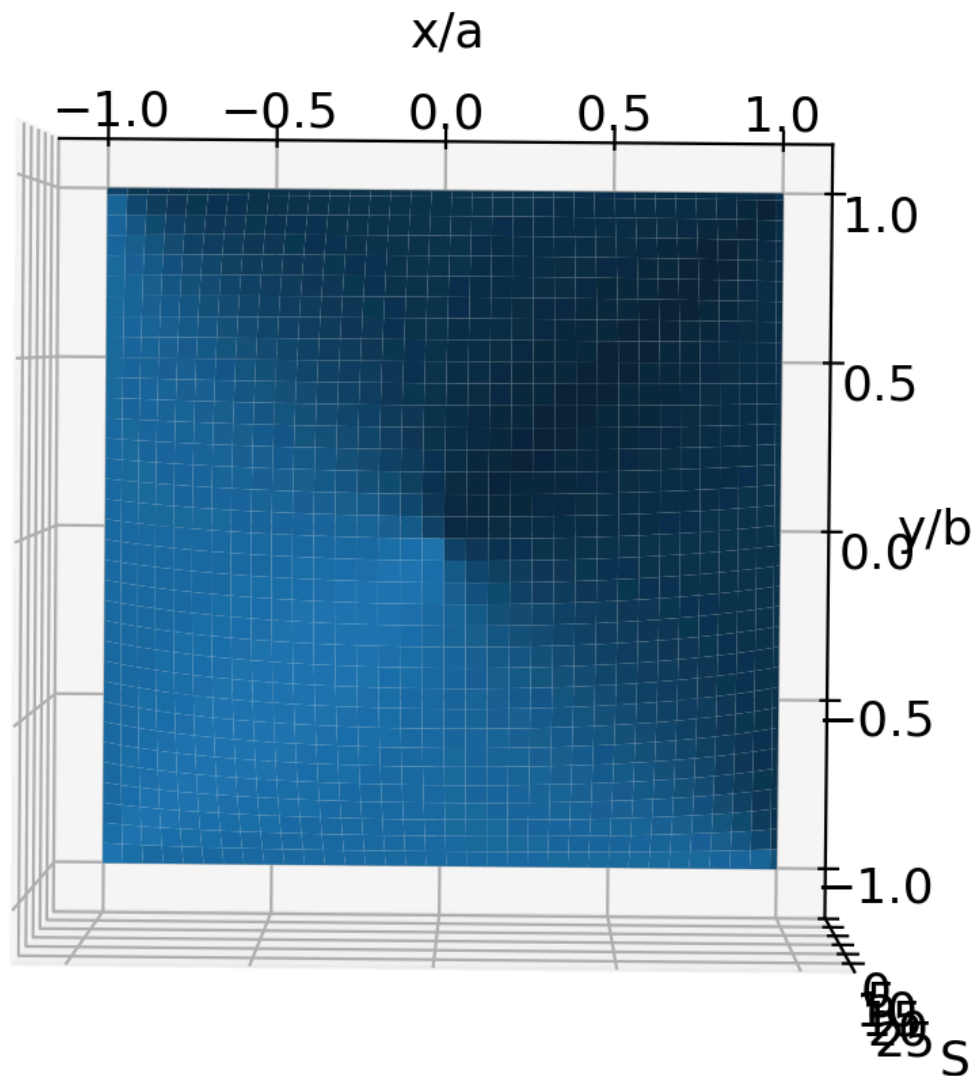
Dimensionless Velocity Profile



Dimensionless Velocity Profile



Dimensionless Velocity Profile



Code For Number 2:

```
import numpy as np
import matplotlib.pyplot as plt

# Dimensionless coordinates
x = np.arange(-1, 1.0001, 0.02) # x/a
y = np.arange(-1, 1.0001, 0.02) # y/b
X, Y = np.meshgrid(x, y)
```

```

AR = 1 # a/b
S = np.zeros_like(X, dtype=float)

for n in range(1, 26):
    for m in range(1, 26):
        term = (
            (-1)**(n + m)
            / (
                (2*n - 1)
                * (2*m - 1)
                * ((2*n - 1)**2 + AR**2 * (2*m - 1)**2)
            )
            * np.cos((2*n - 1) / 2 * np.pi * X)
            * np.cos((2*m - 1) / 2 * np.pi * Y)
        )
        S += term

S = 64 * S

# Plot
fig = plt.figure(figsize=(8, 6))
ax = fig.add_subplot(111, projection='3d')
ax.plot_surface(X, Y, S)

ax.set_title('Dimensionless Velocity Profile')
ax.set_xlabel('x/a')
ax.set_ylabel('y/b')
ax.set_zlabel('S')

plt.show()

```